ONE LAMBDA

A Thermo Fisher Scientific Brand

SeCore™ Sequencing Kits

Instructions for Use

For Research Use Only-Not for Use in Diagnostic Procedures

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SeCore™ Sequencing Kits

Instructions for Use

SeCore™ Sequencing Kits directly identify DNA sequence from target HLA genes. Locus specific DNA amplification is accomplished by cycling an amplification mix, FastStart™ Taq DNA Polymerase and sample genomic DNA. The resulting product is treated with ExoSAP-IT® prior to sequencing to degrade the unincorporated primers and free nucleotides. The nucleotide chronology and resulting HLA subtype is determined by multicolor, fluorescence-based, BigDye™ sequencing. The final reactions are purified by an ethanol precipitation prior to loading. Denatured samples are loaded and results detected on an automated ABI sequencing instrument.

In this manual:

Section	Description	Page
1.	Kit Components	2
2.	Materials, Reagents, and Equipment not Supplied	6
3.	Sample Requirement	7
4.	Locus Specific Genomic DNA Amplification	8
5.	Agarose Gel Electrophoresis	10
6.	Purification of PCR Amplicons (ExoSAP-IT®)	11
7.	Sequencing Reaction	12
8.	Ethanol Precipitation of Sequencing Reaction Products	15
9.	Electrophoresis of Sequencing Reaction Products on a Capillary Sequencer	16
10.	Data Analysis	17
11.	Limitations and Cautions	17
12.	Troubleshooting	18
13.	Symbols	22

1 Kit Components:

Class I Sequencing Kits contents and storage

Upon receipt of the SeCore™ Sequencing Kit, immediately store the components at –10°C to –30°C in a non-frost-free freezer.
To order Class I Sequencing Kits, contact your One Lambda representative.

Part Number		Kit	Pre-PCR Components		Post-PCR Components		
25 tests	**500 tests**	**Class I**	**Amp Mix (locus-specific primers)**	**FastStart® Taq DNA Polymerase (5 U/µL) reagent**	**Exo-SAP IT® reagent**	**Seq Mix (contains sequencing primers, dye terminators, sequencing enzyme)**	**PPT Buffer**
5300925	22009D	A Locus	**25 Test Kit** 1 vial	1 vial	1 vial or bottle	10 vials or bottles total: · Exon 1-Fwd · Exon 1-Rev · Exon2-Fwd · Exon2-Rev · Exon3-Fwd · Exon3-Rev · Exon4-Fwd · Exon4-Rev · Exon5-Fwd · Exon5-Rev	1 vial or bottle
			500 Test Kit 2 bottles				
5311925D	22119D	B Locus	**25 Test Kit** 1 vial	1 vial	1 vial or bottle	9 vials total: · Exon1-Fwd · Exon2-Fwd · Exon2-Rev · Exon3-Fwd · Exon3-Rev · Exon4-Fwd · Exon4-Rev · Exon5-Fwd · Exon5-Rev	1 vial or bottle
			500 Test Kit 2 bottles				
5320925	22209D	C Locus	**25 Test Kit** 1 vial	1 vial	1 vial or bottle	14 vials total: · Exon1-Fwd · Exon1-Rev · Exon2-Fwd · Exon2-Rev · Exon3-Fwd · Exon3-Rev · Exon4-Fwd · Exon4-Rev · Exon5-Fwd · Exon5-Rev · Exon6-Fwd · Exon6-Rev · Exon7-Fwd · Exon7-Rev	1 vial or bottle
			500 Test Kit 2 bottles				

Components are packaged in either 0.5mL or 2.0mL Sarstedt Screw Cap Micro Tubes (72.730.711, 72.609.711) with Sarstedt Screw Caps (65.716.725). Additionally, 500 test pack sizes contain components packaged in either 8.0mL or 15.0mL Nalgene HDPE Narrow Mouth Bottles (342002-9025, 342002-9050).
For vial volumes, refer to the label on each vial.

Class II Sequencing Kits contents and storage

Upon receipt of the SeCore™ Sequencing Kit, immediately store the components at –10°C to –30°C in a non-frost-free freezer. To order Class II Sequencing Kits, contact your One Lambda representative.

Part Number		Kit	Pre-PCR Components		Post-PCR Components		
25 tests	500 tests	Class II	Amp Mix (locus-specific primers)	FastStart® Taq DNA Polymerase (5 U/µL)	ExoSAP-IT® reagent	Seq Mix (contains sequencing primers, dye terminators, sequencing enzyme)	PPT Buffer
5351925	22519D	DPB1 Locus	**25 Test Kit** 2 vials total: · 1 Exon 2 · 1 Exons 3/4 **500 Test Kit** 4 bottles total: · 2 Exon 2 · 2 Exons 3/4	1 vial	1 vial or bottle	8 vials or bottles total: · Exon 2-Fwd · Exon 2-Rev · Exon 3-Fwd · Exon 3-Rev · Exon 4-Fwd · Exon 4-Rev · Codon 8-Fwd · Codon 85-Rev	1 vial or bottle
5341925D	22419D	DQB1 Locus	**25 Test Kit** 2 vials total: · 1 Exon 2 · 1 Exon 3 **500 Test Kit** 4 bottles total: · 2 Exon 2 · 2 Exon 3	1 vial	1 vial or bottle	4 vials or bottles total: · Exon 2-Fwd · Exon 2-Rev · Exon 3-Fwd · Exon 3-Rev	1 vial or bottle
5330925	22309D	DRB1 Locus*	**25 Test Kit** 1 vial total **500 Test Kit** 1 bottle total	1 vial	1 vial or bottle	3 vials or bottles total: · Exon 2-Fwd · Exon 2-Rev · TG Codon 86-Rev	1 vial or bottle
A15572	A15574	DRB1 Locus Exon 2 & 3 Sequencing Kit	**25 Test Kit** 2 vials total: · 1 Exon 2 · 1 Exon 3 **500 Test Kit** 2 Bottles total: · 1 Exon 2 · 1 Exon 3	1 vial	1 vial or bottle	5 vials total: · Exon 2-Fwd · Exon 2-Rev · Exon 3-Fwd · Exon 3-Rev · TG Codon 86-Rev	1 vial or bottle
5331925	Not available	DRB Group (DRB1, DRB3, 4 & 5 Locus)	11 vials †	1 vial	1 vial or bottle	3 vials total: · Group Exon 2-Fwd · Group Exon 2-Rev · TG Codon 86-Rev	1 vial or bottle

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* The DRB1 Locus kit will be identified as the "DRB1 Locus Single Amplification Kit" throughout this document.

†Eleven group-specific Amp Mixes included in the SeCore™ DRB Group Sequencing Kit are: DRB1*01, DRB1*03/11/13/14, DRB1*04, DRB1*07, DRB1*08/12, DRB1*09, DRB1*10, DRB1*15/16, DRB3, DRB4, DRB5

Components are packaged in either 0.5mL or 2.0mL Sarstedt Screw Cap Micro Tubes (72.730.711, 72.609.711) with Sarstedt Screw Caps (65.716.725). Additionally, 500 test pack sizes contain components packaged in either 8.0mL or 15.0mL Nalgene HDPE Narrow Mouth Bottles (342002-9025, 342002-9050).

For vial volumes, refer to the label on each vial.

2 Materials, Reagents, and Equipment not Supplied:

2.1 96-well thermal cycler with heated lid:

2.1.1 SeCore™ Sequencing Kits have been tested on the following thermal cyclers: MJ Research PTC 225 DNA Engine Tetrad, Perkin Elmer Gene Amp 2700, 9600, 9700 and Veriti™. Use of other thermal cyclers will require user validation.

2.2 Automated DNA sequencer and accessories:

2.2.1 SeCore™ Sequencing Kits have been tested on the following DNA sequencers: Applied Biosystems ABI Prism® 3100, 3730 and 3500xL. Use of different DNA sequencers will require user validation.

2.2.2 Reagents and accessories that have been tested with SeCore™ kits:

Instrument	Product Name	Applied Biosystems' Part Number
ABI 3100 Genetic Analyzer (PN 3100-01)	36 cm, 3100 Capillary Array	4315931
	POP-6™ Polymer	4316357
	310 Running Buffer, 10X	402824
	Matrix Standards Kit, BigDye™ Terminator v1.1	4336824
	Hi-Di™ Formamide	4311320
ABI 3730 DNA Analyzer (PN 3730S)	36 cm, 3730 Capilllary Array	4331247
	POP-6™ Polymer	4316357
	3730 Running Buffer, 10X	4335613
	Sequencing Standards, BigDye™ Terminator v1.1	4336799
	Hi-Di™ Formamide	4311320
ABI 3500xL Genetic Analyzer (PN 4440463)	50 cm, 3500xL Capillary Array	4404689
	POP-6™ Polymer for 3500/3500xL Genetic Analyzers	4393717
	Anode Buffer Container (ABC) 3500 Series	4393927
	Cathode Buffer Container (CBC) 3500 Series	4408256
	Sequencing Install Standard, BigDye™ Terminator v1.1	4404314
	Sequencing Install Standard, BigDye™ Terminator v3.1	4404312
	Hi-Di™ Formamide	4440753

2.3 MicroAmp® 96-well Tray/Retainer Set, Applied Biosystems product code 403081.

2.4 MicroAmp® 0.2-ml Reaction Tubes, Reaction Tubes (8 tubes/strip) and Caps (8 caps/strip), Applied Biosystems product code N8011533, N8010580 and N8010535.

2.5 MicroAmp® Optical 96-Well Reaction Plate and 9600 Full Plate Cover, Applied Biosystems product code N8010560 and N8010550.

2.6 Table top centrifuge with plate adapters for 96-well plates. The centrifuge must reach a force of 2500 x g.

2.7 Pipettors and tips: 1-10 µl, 10-200 µl, 100-1000 µl.

2.8 Dispensing electronic pipettors: capable of dispensing 1- 125 µl aliquots.

2.9 Multichannel (8 or 12 channels) pipettors: 1-100 µl adjustable volume.

2.10 Agarose gel electrophoresis apparatus.

2.11 CoolSafe System to fit 0.2 ml tubes or ice, Diversified Biotech product code CSAF-1000

2.12 TE-4 (10mM Tris, pH 8.3, 0.1mM EDTA), Teknova product code T0223

2.13 Absolute Ethanol, Sigma Aldrich product code E7023

2.14 Molecular Biology grade Agarose, Invitrogen product code 75000500

2.15 Ethidium Bromide, Amresco product code X328-10mL

2.16 PCR Marker suitable to cover range of 300 – 1300 bp, Invitrogen product code 74601250 (recommended) and Loading Dye without Bromophenol Blue or Xylene Cyanol

2.17 Sequencer software:

2.17.1 For use with ABI 3100 and 3730

- ABI Prism® Data Collection Software v1.1 or higher (Win NT)
- Sequencing Analysis Software v3.7 or higher (Win NT)

2.17.2 For use with ABI 3500xL

- 3500xL Data Collection Software v1.0 (Windows® Vista SP1)

2.18 HLA Analysis Software

2.18.1 One Lambda Inc. uTYPE™ HLA Sequence Analysis Software product code 539991

3 Sample Requirement:

3.1 Purified DNA with an $OD_{260/280}$ between 1.7 and 1.9.

Note: The Qiagen QIAamp™ Kit has been successfully used for this procedure.

Note: Blood samples should be collected in ACD or EDTA anticoagulated tubes. Do not use Heparinized samples.

IMPORTANT! Perform DNA extraction according to DNA extraction kit manufacturer instructions. Phenol, Ethanol and SDS may cause inhibition of the PCR reaction.

3.2 To prepare a working solution of the purified DNA at 15-30 ng/µl, dilute the DNA in TE-4. DNA can also be diluted in water for immediate use. DNA aliquoted in water should be discarded after use. TE-4 is recommended for long term storage.

Note: Change pipet tips in between the pipetting of each DNA sample and each different mix or reagent to prevent cross contamination. The same pipet tip may be used to dispense the same mix or reagent into multiple tubes provided the tip does not come into contact with genomic DNA or PCR product. If there is any question that this may have occurred, change the pipet tip to prevent contamination.

4 Locus Specific Genomic DNA Amplification:

The Amp Mix of the SeCore™ Sequencing Kits does not contain the FastStart™ DNA Polymerase. To prepare the master mix for genomic DNA amplification, combine the locus specific Amp Mix with FastStart™ Taq DNA Polymerase in a clean tube, vortex to mix and centrifuge briefly to bring the contents of the tube to the bottom.

Note: For 25 Test kits, it is recommended that a master mix for a minimum of 5 reactions be prepared at one time to aid in the accuracy of pipetting small volumes of enzyme.

Note: For 500 test kits, prepare a master mix for a minimum of 20 samples to limit the amount of inadvertent enzyme loss during pipetting. Kit reagent volumes are sufficient for 25 sets of such reactions.

IMPORTANT! Due to the viscous nature of FastStart Taq Polymerase, care should be taken to remove excess Taq that remains on the pipette tip before dispensing the enzyme into the master mix. Taq should be pipetted from the top of the vial and the user should avoid submerging the tip into the enzyme. Failure to remove excess Taq will result in a shortage of the enzyme.

The amplification reactions can be prepared in individual tubes or strip tubes. MicroAmp® tubes and strips have been tested for use with this procedure. Use of plasticware from other manufacturers will require user validation.

4.1 Preparation of Class I amplification reactions:

Locus	Amp Mix Sets	Notes
A, B and C	1	The amplification set up will depend on the number of sequencing mixes that are going to being tested. Eight sequencing mixes or less will be tested: • Set up one reaction for each sample plus one negative control per group of samples being amplified. More than eight sequencing mixes will be tested: • Set up two reactions for each sample plus one negative control per group of samples being amplified.

4.1.1 For each amplification reaction, combine 19.8 µl of Amp Mix with 0.2 µl of FastStart™ Taq DNA Polymerase. Pulse vortex 2-3 times. To make a master

mix for multiple reactions, use the table below as an example of how to calculate volumes of Amp Mix and FastStart™ Taq.

Class I	n = Number of Reactions	
	n = 5	n = 20
Amp Mix 19.8 µl per reaction	99 µl	396 µl
FastStart™ Taq 0.2 µl per reaction	1.0 µl	4.0 µl

4.1.2 Add 5 µl of each sample DNA (15-30 ng/µl) to a reaction tube. Add 5 µl of sterile water to the negative control reaction.

4.1.3 Add 20 µl of the mixture of Amp Mix and FastStart™ Taq to each tube.

4.1.4 Cover the tubes and centrifuge briefly.

4.1.5 Place the tubes in the thermal cycler. Start the PCR according to the amplification profile described in Section 4.3.

Note: Section 4.1.4 through 4.1.8 must be performed efficiently to minimize the time between mixing of Taq with Amp Mix and the initiation of thermal cycling.

4.2 Preparation of Class II amplification reactions:

4.2.1 The DRB1 Locus Single Amplification Kit consists of one Amp Mix. Set up one reaction for each sample plus one negative control per group of samples being amplified.

4.2.2 The DRB1 Locus Exon 2 & 3 Kit consists of two Amp Mixes. Set up two reactions for each sample plus one negative control per group of samples being amplified.

4.2.3 The DRB **Group** kit consists of 11 sets of Amp Mix. It is designed to amplify the 11 major DRB1/3/4/5 groups separately for better resolution and ease of analysis. All 11 amplification reactions will be performed for samples of unknown typing. Set up 11 reactions for each sample. Set up one negative control for each set of Amp Mix.

Note: When a low resolution HLA typing of the sample is known, the user may choose the necessary groups to be amplified for high resolution DRB1 sequencing.

4.2.4 The DPB1 kit consists of 2 vials of Amp mix. Both amplification reactions must be done for samples of unknown typing. Set up one negative control for each set of Amp Mix.

4.2.5 The DQB1 Locus kit contains two sets of Amp Mix. Both amplifications must be done for each sample. Set up two reactions for each sample. Set up one negative control for each set of Amp Mix.

4.2.6 For each amplification reaction, combine 22.8 µl of Amp Mix with 0.2 µl of FastStart™ Taq DNA Polymerase. Pulse vortex 2-3 times.To make a master mix for multiple reactions, use table below as an example of how to calculate volumes of Amp Mix and FastStart™ Taq.

Class II	n = Number of Sample	
	n = 5	n = 20
Amp Mix 22.8 µl per reaction	114 µl	456 µl
FastStart™ Taq 0.2 µl per reaction	1.0 µl	4.0 µl

4.2.7 Add 2 µl of each sample DNA (15-30 ng/µl) to a reaction tube or well. Add 2 µl of sterile water to the negative control reaction

4.2.8 Add 23 µl of the mixture of Amp Mix and FastStart™ Taq to each tube.

4.2.9 Cover the tubes and centrifuge briefly.

4.2.10 Place the tubes in the thermal cyler, start the PCR according to the amplification profile described in Section 4.3.

Note: Section 4.2.4 through 4.2.7 must be performed efficiently to minimize the time between mixing of Taq with Amp Mix and the initiation of thermal cycling.

4.3 Genomic DNA amplification profile:

Note: The amplification profile is identical for Class I and Class II amplifications.

Step	**Number of Cycles**	**Temperature**	**Time**
Soak	1	95°C	4 min
Cycle	35	95°C	20 sec
		63°C	20 sec
		72°C	40 sec
Extension	1	72°C	5 min
Soak	1	4°C	Infinite*

The total PCR time is ~1.5 hours.

*Remove in time to proceed to the next step

Note: If sections 5 and 6 are not performed immediately, PCR products can be stored at 4°C. However, the Purification of the PCR Amplicons (ExoSAP-IT®) should be done within 18 hours.

5 Agarose Gel Electrophoresis:

5.1 Confirm the PCR products by agarose (2.0%) gel electrophoresis. Load agarose gel with 5µl of PCR product plus loading dye and run the gel according to your laboratory protocol. Run the gel for a sufficient amount of time to see separation of the Class I and DR Group Specific bands. Amplicons of the correct size should be greater than 35 ng per band when 5 µl are loaded per well. When stained with ethidium bromide (250 ng/ml Agarose) and visualized under UV light, the product band should be equivalent or stronger in intensity when compared to any band of the recommended marker except the more intense 500 bp band.

5.2 Expected products for each locus specific amplification

Kit	**Product Size (bp)**	**Internal Control Size (bp)**
A Locus	~1200 and ~990	None
B Locus Single Amp	~1400 and ~950	None
C Locus	~1375 and ~1600	None
DRB1 Locus Single Amplification Kit	~300	None
DRB1 Locus Exon 2 & 3 Sequencing Kit	Exon 2: ~500-850 Exon 3: ~450	None
DRB Group	~300	~600
DQB1 Locus	~350 and ~375	None
DPB1 Locus	~300 and ~1000	None

Note: A weak band larger than the 300bp band may be seen for some DRB1 Locus Single Amplification Kit amplification products. The sequencing result is not affected by the presence of the doublet.

5.3 For confirming the DRB group specific amplifications if all groups are amplified, all reactions should be checked on an agarose gel. Sequencing reactions should only be set up for the positive 300bp reactions.

6 Purification of PCR Amplicons (ExoSAP-IT®):

Treating the PCR amplicons with ExoSAP-IT® will degrade unincorporated primers and hydrolyze dNTPs.

IMPORTANT! Follow the *Taq* guidelines listed above for pipetting viscous enzymes when adding ExoSAP-IT®

6.1 If you did not perform gel electrophoresis, (see Agarose Gel Electrophoresis on page 9), remove and discard 5 µL from each amplified DNA product.

6.2 Add 4 µl of ExoSAP-IT® to the amplification tubes.

6.2.1 For the DRB group kit, add ExoSAP-IT® only to the tubes having positive 300bp products

6.3 Cover the tubes and centrifuge ~5 seconds to bring contents to the bottom of the tube.

6.4 Vortex the reaction mixtures thoroughly (~10 seconds)

Note: Make sure that ExoSAP-IT® is mixed well with PCR products. This is critical to the success of the purification of PCR amplicons.

6.5 Centrifuge briefly to pull the contents to the bottom of the tubes or wells.

6.6 Place the tubes or plate in a thermal cycler and run the following profile:

Step	**Number of Cycles**	**Temperature**	**Time**
Cycle	1	37°C	20 min
		80°C	20 min
Soak	1	4°C	infinite

Note: ExoSap-IT® treated amplicons can be stored at –20°C for up to two weeks.

7 Sequencing Reaction:

7.1 Sequencing reactions for each locus:

A Locus	B Locus	C Locus
Exon1 forward	Exon1 forward	Exon1 forward
Exon1 reverse	Exon2 forward	Exon1 reverse
Exon2 forward	Exon2 reverse	Exon2 forward
Exon2 reverse	Exon3 forward	Exon2 reverse
Exon3 forward	Exon3 reverse	Exon3 forward
Exon3 reverse	Exon4 forward	Exon3 reverse
Exon4 forward	Exon4 reverse	Exon4 forward
Exon4 reverse	Exon5 forward	Exon4 reverse
Exon5 forward	Exon5 reverse	Exon5 forward
Exon5 reverse		Exon5 reverse
		Exon6 forward
		Exon6 reverse
		Exon7 forward
		Exon7 reverse

7.1.1 Set up forward and reverse sequencing reactions for exons 2, 3, and 4 for all samples. Sequencing reactions for additional exons can be set up concurrent to exons 2, 3, and 4 or within 2 weeks of amplification date if purified reactions are stored at -20°C.

DRB1 Locus Single Amplification Kit	**DRB1 Locus Exon 2 & 3 Sequencing Kit**	**DRB Group**	**DQB1 Locus**	**DPB1 Locus**
Exon2 forward Exon2 reverse Codon 86 reverse	Exon2 forward Exon2 reverse Exon3 forward Exon3 reverse Codon 86 reverse	Exon2 forward Exon2 reverse Codon 86 reverse	Exon2 forward Exon2 reverse Exon3 forward Exon3 reverse	Exon2 forward Exon2 reverse Exon3 forward Exon3 reverse Exon4 forward Exon4 reverse Codon 8 forward Codon 85 reverse

7.1.2 For each positive DRB group specific PCR reaction, set up 2 sequencing reactions. The minimum number of sequencing reactions is 2, when a sample has only one positive amplification. The maximum number of sequencing reactions is 8, when a sample has two positive amplifications of DRB1 locus and two positive amplifications of DRB3/4/5 loci. It is optional to set up the Codon 86 reaction (see Section 12).

7.1.3 It is very important to verify that the appropriate sequencing mix and the treated amplicon are placed in the correct tube or well.

For the DQB1 Locus kit, set up forward and reverse sequencing reactions for exons 2 and 3 for both of the positive amplification reactions. Exon 2 sequencing mix should be used with exon 2 amplicon and exon 3 mix should be used with the exon 3 amplicon, resulting in a total of 4 reactions per sample.

For the DPB1 Two Amp kit, set up forward and reverse sequencing reactions for exons 2-4. Exon 2 sequencing mix should be used with exon 2 amplicon. Exon 3 and 4 sequencing mixes should be used with the exon 3/4 amplicon.

For the DRB1 Exon 2 & 3 Sequencing Kit, set up forward and reverse sequencing reactions for exons 2 and 3. Exon 2 sequencing mix should be used with exon 2 amplicon and exon 3 mix should be used with the exon 3 amplicon. For codon 86 sequencing, see page 15.

7.1.4 Sample layout will depend on instrument configuration, i.e. number of capillaries. See uTYPE™ Sequencing Analysis software instructions for use for sample naming recommendations.

7.1.4.1 For 16 capillary users, the following Optical 96- well Plate sample layout is suggested:

	1	2	3	4	5	6	7	8	9	10	11	12
A	S1F	S1R										
B	S2F	S2R										
C												
D												
E												
F												
G												
H												

Note: S1F and S2F stand for forward reactions of sample 1 and 2, whereas S1R and S2R stands for reverse reactions of sample 1 and 2.

7.2 SeCore™ DR Codon 86 sequencing

Ambiguous DRB1 typing results

If DRB1 typing results are ambiguous, examine the Codon 86 associated nucleotides. Two consecutive positions with a KK sequence (K= G/T) indicate a Codon 86 related ambiguity. These can arise when using the DRB1 Locus Single Amplification, DRB1 Locus Exon 2 & 3 Sequencing Kits or the DRB1 amplicons from the DRB Group Sequencing Kit:

To resolve the ambiguity, prepare a new sequencing reaction using the TG Codon 86 Rev Seq Mix.

DRB1 gene groups

DRB1 genes can be divided into two groups based on the sequence at Codon 86:

- The group with the sequence GTG.

or

- The group with GGT at this position.

Using the target specific, GTG, sequencing primer (TG Codon 86 Rev Seq Mix) allows for separate sequencing of one of the two sample alleles, to resolve the Codon 86 associated ambiguity. The result will be the anti-sense sequence of one allele of the sample allele pair.

DR Codon 86 sequencing can be set up with or after DR sequencing.

Procedure for Codon 86 sequencing

1. Set up the sequencing reaction using the ExoSAP-IT®treated amplicons of the DRB1 Locus Single Amplifcation and DRB1 Locus Exon 2 & 3 Sequencing Kits and DRB Group Sequencing Kit. See Prepare sequencing reactions on page 13.
2. Perform the procedures in Ethanol Precipitation of Sequencing Reaction Products on page 14 and Electrophoresis of Sequencing Reaction Products on a Capillary Sequencer on page 15.
3. Analyze the data together with the data obtained using the DRB1 Locus Single Amplification and DRB1 Locus Exon 2 & 3 Kits or DRB Group Kit.

7.3 SeCore™ DPB1 Codons 8 and 85 sequencing

Ambiguous DPB1 typing results

If DPB1 typing results are ambiguous, examine the sequence at Codon 8 and Codon 85. Distinguish ambiguous results by using the nucleotide sequence at Codon 8, Codon 85, or by using both codons. Use the Codon 8 CTT forward sequencing primer (Z39) and Codon 85 GAG reverse sequencing primer (Z38) to identify the sequence at these locations. Set up DP Codon 8 and Codon 85 sequencing with or after DP sequencing.

Procedure for Codon 8 and 85 sequencing

1. Set up the sequencing reaction using the ExoSAP-IT® treated exon 2 amplicon. See Prepare sequencing reactions on page 13.
2. Perform the procedures in Ethanol Precipitation of Sequencing Reaction Products on page 14 and Electrophoresis of Sequencing Reaction Products on a Capillary Sequencer on page 15.
3. Analyze the data together with the data obtained using the SeCore™ DPB1 Sequencing Kit.

7.4 Preparation of sequencing reactions:

Note: Sequencing reaction mixtures should be kept as cold as possible, by either using the recommended CoolSafe system or ice.

7.4.1 For Class II reactions only, add 40 µl of ultra pure water to ExoSAP-IT® treated amplicons and mix well.

7.4.2 For Class I and Class II reactions, add 2 µl of ExoSAP-IT® treated PCR amplicons (or Class II amplicon dilutions from step 7.2.1) to the appropriate tube or well.

7.4.3 Add 8 µl of each locus specific sequencing mix to the appropriate tube or well.

7.4.4 Cover the tubes or plate, vortex briefly to mix and centrifuge the tubes or plate to bring contents to the bottom of the tubes or plate.

7.4.5 Place the tubes or plate in the thermal cycler and run the profile described in Section 7.3.

7.5 Sequencing profile:

7.5.1 Start the run. When the thermal cycler block is >80 °C, place the tubes or plate in the thermal cycler, then close the lid.

Note: The sequencing profile is identical for Class I and Class II sequencing reactions.

Step	**Number of Cycles**	**Temperature**	**Time**
Cycle	25	95°C	20 sec
		50°C	15 sec
		60°C	60 sec
Soak	1	4°C	Infinite*

The total reaction time is ~1.5 hrs.

*Remove in time to proceed to the next step

Note: Continue with the ethanol precipitation of the sequencing reaction products within 24 hours (see instructions of Section 8).

8 Ethanol Precipitation of Sequencing Reaction Products:

After cycle sequencing, an ethanol precipitation is performed to remove excess terminators. The instructions are identical for Class I and Class II sequencing reactions.

8.1 Add 2 µl of Sequencing PPT Buffer to each sequencing reaction mixture.

Note: Use of a single channel pipette or single channel repeater pipette is recommended for this step. Dispensing PPT Buffer into a reagent trough or boat to aid in multi channel pipetting can result in significant reagent loss.

8.2 Centrifuge briefly to ensure all contents are mixed and at the bottom of the well.

8.3 Add 40 µl of 100% ethanol to each mixture.

8.4 Cover the plate and vortex well for a minimum of 60 seconds.

Note: Make sure that the contents of all tubes are being mixed thoroughly.

8.5 Centrifuge the plate in a centrifuge fitted with a plate-adaptor for 30 minutes at 2,000 x g or greater.

8.6 Remove cover and invert onto paper towel. Centrifuge briefly (1 min. at 500 x g) in the inverted position to remove as much liquid as possible.

8.7 Add 100 µl of 80% ethanol to the pellets. Do **NOT** vortex the plate.
Note: Prepare fresh 80% ethanol daily.

8.8 Centrifuge the plate for 5 min at 2,000 x g or greater.

8.9 Remove the supernatant by an inverted spin as in step 8.6.

9 Electrophoresis of Sequencing Reaction Products on a Capillary Sequencer:

9.1 Preparation of Loading Samples

9.1.1 Add 15 µl of Hi-Di™ Formamide (Applied Biosystems product code 4311320) to each DNA pellet.

9.1.2 Cover the plate and briefly centrifuge to bring contents to the bottom of the plate wells

9.1.3 Denature the samples at 95°C for 2 minutes in a thermal cycler.

9.1.4 Centrifuge briefly to remove any air bubbles in the samples.

Note: If air bubbles get into the capillary, they will cause damage to the capillary.

9.1.5 Place the plate in the sequencer autosampler.

9.2 Suggested Electrophoresis Conditions on an ABI Capillary Sequencers:

Instrument	Parameters	POP-6 Settings
ABI 3100	Run Module	RapidSeq36_POP6
	Injection Time	10 seconds

	Run Time	1800 seconds
ABI 3730	Run Module	StdSeq36
	Injection Time	5 seconds
	Run Time	1800 seconds
ABI 3500xL	Run Module	StdSeq50
	Injection Time	Default
	Run Time	3780 seconds

Note: The first time BigDye™ 1.1 Terminators are used, the user must run matrix standards. Follow the package instructions for the BigDye™ Terminator Matrix Standard (product code 4336824).

Parameters and settings for other instrument configurations may require optimization.

10 Data Analysis:

10.1 Use Sequencing Analysis software to process collected raw data and create sequence files.

10.2 Use One Lambda Inc. uTYPE™ HLA Sequence Analysis Software to process the sequence files and create HLA typing reports.

Note: It is strongly recommended that the most recent IMGT allele database is implemented for sequence data analysis.

Note: For users of the DRB1 Locus Exon 2 & 3 Sequencing Kit, the appropriate changes need to be made to the Exon Trim Options settings within the uTYPE™ HLA sequence analysis software. With the Preferences menu, click the Exon Trim Options tab and change the DRB1 exon 2 and 3 trim values from "Default" to "Full Exon".

11 Limitations and Cautions:

11.1 Before implementing the SeCore™ HLA Sequencing Kit in your laboratory, perform quality assurance and quality control for sequencing based typing methods using known molecular typed samples. Such samples can be obtained from the International Workshop Reference Cell Panel and the UCLA DNA Reference Panel.

11.2 Every attempt has been made to validate all amplification primers used in SeCore™ HLA Sequencing Kits with DNA samples typed by molecular methods. Due to the limited access to such reference materials, some rare alleles may not have been tested with these kits.

11.3 Genomic DNA PCR reactions may be set up on ice or the CoolSafe system if desired.

11.4 Some DR alleles are amplified under other groups of DRB Group Sequencing Kit. The list below may not be complete.

Alleles	Amplified by Amp Mix of
DRB1*0820	DRB1*03/11/13/14
DRB1*1122 DRB1*1410	DRB1*04
DRB1*1105 DRB1*1317 DRB1*1404 DRB1*1411 DRB1*1415 DRB1*1428 DRB1*1431	DRB1*08/12
DRB1*1130 DRB1*1446	DRB3

11.5 The Amp Mix of DRB1 Locus Single Amplification Sequencing Kit does not amplify alleles DRB1*1130 and DRB1*1446. However, these alleles can be amplified by the DRB3 Amp Mix of the DRB Group Sequencing Kit.

11.6 A weak band larger than the 300bp band may be seen for some DRB1 Locus Single Amplification Kit amplification products. The sequencing result is not affected by the presence of the doublet.

11.7 The possible ambiguities of all loci can be found at www.ebi.ac.uk/imgt/hla/ambig.html.

11.8 To ensure the best performance of SeCore™ HLA Sequencing Kits, use the products with the materials, reagents, and equipment recommended in Section 2 of this document. Use of other materials must be validated by the user.

11.9 The A locus Exon 4-Reverse sequencing primer contains a mismatch towards A*32 and A*74 samples. Additionally, this mismatch applies to the A*02:65 allele. Samples containing these alleles may exhibit poor heterozygous peak balance in the Exon 4-Reverse sequence data. The Exon 4-Forward sequence data will be unaffected.

11.10 In rare cases, alleles with unknown variants or newly-defined sequence at amplification primer binding sites may affect amplification efficiency and result in incorrect typing results.

12 Troubleshooting:

Problems	Possible Causes	Solutions
Absence of PCR product or weak PCR product	FastStart Taq™ was not added to the amplification mix or not mixed properly when added	Repeat amplification paying attention to the addition and mixing of taq with the amplification mix. Use a tube that is appropriate size for the amount of mix.

Problems	**Possible Causes**	**Solutions**
	Thermal cycler failure	Check the cycler run history. Perform maintenance or reprogram the thermal cycler.
	DNA concentration not optimal	Re-quantitate DNA and adjust to 15-30ng/µl. Note: In the event that the sample concentration is below the recommended range and a weak or no amplification occurs, sequence the sample. Acceptable sequence and typing may be achievable.
	Poor quality or degraded DNA	Run DNA on a 1% agarose gel to evaluate quality. Intact genomic DNA should be >3000bp. Re-amplify the sample using 0.3µl of Taq per test.
	The visibility of the product bands on the agarose gel are masked by the dye in the loading buffer	Repeat gel electrophoresis using a different gel loading buffer. ClearBand™ Sample Loading Buffer, MTR Scientific product code CB5, is suggested.
Presence of extra amplification products (With the exception of the DRB1 product).	Incorrect thermal cycling program was used	Repeat amplification and confirm the correct program was used.
	Amplification mix, DNA or work area is contaminated	De-contaminate the work area or re-extract DNA. Repeat amplification.
Excessive background (baseline noise)	ExoSAP-IT® not added to the amplicons prior to sequencing	Add ExoSAP-IT® to the PCR product.
	ExoSAP-IT® not thoroughly combined with the PCR product	Be sure to vortex thoroughly after the addition of ExoSAP-IT®.
	ExoSAP-IT® reagent was over-vortexed	Be sure to vortex briefly but thoroughly after the addition of ExoSAP-IT®.
	Signal strength too high	See "Excessive Signal Strength" below.
	Poor or incorrect matrix	ABI 3100, 3730: Repeat the spectral calibration and reinject samples.
	Poor injection (ABI 3100)	Reinject samples.

Problems	**Possible Causes**	**Solutions**
	ABI 3100, 3730: Injection time set too high	ABI 3100, 3730: Reduce injection time and reinject. (Samples of poor quality may have lower signal strengths but may still be analyzed and typed. Some samples may have signals that are over 1500 and will not have excess background.)
	Poor sequencing reaction due to error in pipetting	Be sure that both the ExoSAP-IT® treated PCR product and the correct sequencing mix are added and combined.
	Incorrect mobility file was chosen- peaks will be shifted or will be on top of each other	Choose correct mobility file.
	Excess 100% EtOH was added	Resequence. Check to be sure pipet is set at correct volume.
	Ethanol evaporation during precipitation	Prepare new sequencing reactions. Purify the sequencing extension products away from direct air flow (for example, exhaust from a centrifuge or other equipment), which may cause ethanol evaporation.
Weak Signal	Sequencing reactions were not vortexed thoroughly after the addition of the Sequencing PPT Buffer and EtOH	Repeat sequencing reactions. Vortex thoroughly (minimum of 60 seconds) after the addition of EtOH. Be sure all reactions are mixing.
	Injection time needs to be increased	Repeat sequencing reaction(s) and increase injection time.
	Precipitate is present in PPT buffer	Vortex until particles are dissolved and buffer is clear.
Excessive Dye Blobs	The sequencing PPT buffer was not added to the sequencing reactions prior to the addition of EtOH	Repeat sequencing reaction. Add the sequencing ppt buffer prior to the addition of EtOH.
	Failure to wash the sequencing reactions with 70% - 80% EtOH	Repeat sequencing reactions without omitting the 70% - 80% EtOH wash step.

Problems	**Possible Causes**	**Solutions**
	Poor sequencing reaction due to error in pipetting or weak amplification product	Be sure that both the ExoSAP-IT® treated PCR product and the correct sequencing mix are added and combined. In the case of a weak amplification, confirm the intensity of the amplification product by running an agarose gel.
	Failure to remove all of the remaining EtOH during the precipitation	Repeat sequencing reactions. The centrifuge step of 500 x g is very important in removing all remaining ethanol in the reaction tubes.
	EtOH used in wash step was too dilute	Repeat sequencing reactions and confirm that 70% - 80% EtOH is being used for the wash step.
Excessive Signal Strength	ABI 3100, 3730: Injection time set too high	ABI 3100, 3730: Reduce injection time and reinject. Samples of poor quality may have lower signal strengths but may still be analyzed and typed. Some samples may have signals that are 2000-3000 and will not have excess background.
Weak signal from short fragments Longer fragments show a stronger signal	Did not use fresh 80% ethanol	Prepare fresh 80% ethanol daily.
Strong signal from short fragments Longer fragments show a weaker signal	Over amplification of short fragments during purification	Repeat sequencing reactions. Reduce the amount of ethanol stepwise during the wash step down to a minimum concentration of 70%
Random sequence failures	Poor sequencing reaction due to error in pipetting	Be sure that both the ExoSAP-IT® treated PCR product and the correct sequencing mix are added and combined.

13. Symbols

EXPLANATION OF SYMBOLS (reference EN ISO 15223-1: Medical devices – Symbols to be used with medical device labels, labeling and information to be supplied)

Symbol	Definition
ISO 7000 Reg No. 0632	Temperature Limitation (Range)
ISO 7000 Reg No. 0534	Lower Temperature Limitation
ISO 7000 Reg No. 1641	Upper Temperature Limitation
ISO 7000 Reg No 2607	Use By
ISO 7000 Reg No. 1641	Consult Instructions for Use
REF *ISO 7000 Reg No. 2493*	Catalog Number
LOT *ISO 7000 Reg No 2492*	Batch Code
ISO 7000 Reg No. 3082	Manufacturer
ISO 7000 Reg No. 0659	Warning: Product May Contain Biohazardous Material
Σ *ISO 7000 Reg No.0518*	Number of Tests
ISO 7000 Reg No 2497	Date of Manufacture
EC REP	Authorized Representative in the European Community
ISO 7000 Reg No. 0434A	Warning: Attention, See Instructions for Use

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Revision 04

REF	
22009D	SeCore™ A Locus Sequencing Kit (RUO)
22119D	SeCore™ B Locus Sequencing Kit, Single Amplification (RUO)
22209D	SeCore™ C Locus Sequencing Kit (RUO)
22309D	SeCore™ DRB1 Locus Sequencing Kit (RUO)
A15574	SeCore™ DRB1 Locus Exon 2 & 3 Sequencing Kit (RUO)
22419D	SeCore™ DQB1 Locus Sequencing Kit (RUO)
22519D	SeCore™ DPB1 Locus Sequencing Kit (RUO)
5300925	SeCore™ A Locus Sequencing Kit (RUO)
5311925D	SeCore™ B Locus Sequencing Kit, Single Amplification System
5320925	SeCore™ C Locus Sequencing Kit (RUO)
5330925	SeCore™ DRB1 Locus Sequencing Kit (RUO)
A15572	SeCore™ DRB1 Locus Exon 2 & 3 Sequencing Kit (RUO)
5331925	SeCore™ DRB Group Sequencing Kit (RUO)
5341925D	SeCore™ DQB1 Locus 2AMP (RUO)
5351925	SeCore™ DPB1 Locus Sequencing Kit, Group Specific Set (RUO)

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